



KEMENTERIAN PENDIDIKAN TINGGI, SAINS DAN TEKNOLOGI
UNIVERSITAS PADJADJARAN

Jalan Dipati Ukur No. 35 Bandung 40132
Jalan Ir. Soekarno Km. 21 Jatinangor, Sumedang 45363
Telepon (022) 84288888 Laman: www.unpad.ac.id, Email: humas@unpad.ac.id

**PERJANJIAN PELAKSANAAN KEGIATAN
HIBAH ARTIKEL *REVIEW* BATCH 2 EQUITY-WCU
UNIVERSITAS PADJADJARAN TAHUN ANGGARAN 2025**

Nomor : 5652/UN6.3.1/PT.00/2025

Pada hari ini **Senin** tanggal **Dua puluh Dua** bulan **Desember** tahun **dua ribu dua puluh lima**, kami yang bertandatangan di bawah ini:

- 1. Prof. dr. Nur Atik, M.Kes., Ph.D.** : Direktur Riset, Hilirisasi dan Pengabdian pada Masyarakat Universitas Padjadjaran dalam hal ini bertindak untuk dan atas nama Universitas Padjadjaran sesuai dengan Keputusan Rektor tentang Pengangkatan Direktur Nomor 2248/UN6.RKT/Kep/HK/2024 tanggal 18 November 2024 yang bekedudukan di Jalan Ir. Soekarno Km. 21 Jatinangor Sumedang, selanjutnya disebut **PIHAK PERTAMA**.
- 2. Dr. Restu Arisanti, S.Si., M.Si.** : Tenaga Pendidik Fakultas Matematika dan Ilmu Pengetahuan Alam Universitas Padjadjaran Jalan Ir. Soekarno Km. 21 Jatinangor Sumedang, selanjutnya disebut **PIHAK KEDUA**.

Perjanjian penugasan ini berdasarkan pada:

1. Keputusan Rektor Universitas Padjadjaran Nomor 1653/UN6.RKT/Kep/HK/2025 Tentang Pedoman Hibah Riset Equity - WCU Tahun 2025 Universitas Padjadjaran, tanggal 1 September 2025;
2. Keputusan Rektor Universitas Padjadjaran Nomor 1659/UN6.RKT/Kep/HK/2025 Tentang Penetapan Besaran Dana Hibah Riset Equity – WCU Universitas Padjadjaran, tanggal 2 September 2025;
3. Keputusan Rektor Universitas Padjadjaran Nomor 2315/UN6.RKT/Kep/HK/2025 Tentang Penetapan Penerima Hibah Riset Equity - WCU Universitas Padjadjaran Skema RKI-AR Batch 2 Tahun Anggaran 2025, tanggal 19 Desember 2025.

PIHAK PERTAMA dan **PIHAK KEDUA** secara bersama-sama bersepakat mengikatkan diri dalam suatu Perjanjian Pelaksanaan “Penulisan Artikel *Review* Batch 2 Equity - WCU Universitas Padjadjaran Tahun Anggaran 2025”, selanjutnya disebut Hibah Penulisan Artikel *Review* Unpad T.A. 2025, dengan ketentuan dan syarat-syarat yang diatur dalam pasal-pasal berikut:

**PASAL 1
TUJUAN**

Perjanjian Pelaksanaan Hibah Penulisan Artikel *Review* Batch 2 Equity - WCU Unpad T.A. 2025 ini dibuat dalam rangka pemberian tugas kegiatan riset yang diselenggarakan oleh Universitas Padjadjaran.

PASAL 2 LINGKUP KEGIATAN

- (1) **PIHAK PERTAMA** memberi tugas kepada **PIHAK KEDUA** untuk melaksanakan Hibah Penulisan Artikel *Review Batch 2 Equity - WCU T.A. 2025*, dengan judul “*Methodological Developments of Generalized Linear Mixed Models: A Systematic Review with Applications in Tuberculosis Epidemiology*”.
- (2) **PIHAK KEDUA** bertanggungjawab penuh atas pelaksanaan, administrasi dan keuangan atas pekerjaan sebagaimana dimaksud pada ayat (1) dan berkewajiban menyimpan semua bukti-bukti pengeluaran, serta mengirimkan *output* riset yang telah disepakati dalam perjanjian ini kepada **PIHAK PERTAMA**.
- (3) Pendanaan pelaksanaan Hibah Penulisan Artikel *Review Batch 2 Equity - WCU Unpad T.A. 2025* sebagaimana dimaksud pada ayat (1) dibebankan pada Anggaran Universitas Padjadjaran PTN-BH.

PASAL 3 PEMBIAYAAN PENELITIAN

- (1) **PIHAK PERTAMA** memberikan dana Total Pelaksanaan kegiatan sebagaimana dimaksud dalam pasal 1 sebesar **Rp. 25.000.000,- (Dua puluh lima juta rupiah)**.
- (2) Dana Riset sebagaimana dimaksud pada ayat (1) dibayarkan oleh **PIHAK PERTAMA** kepada **PIHAK KEDUA** secara bertahap melalui rekening **PIHAK KEDUA**, dengan ketentuan sebagai berikut:
 - a. Pembayaran Tahap Pertama sebesar 40% dari total dana yaitu $40\% \times \text{Rp. 25.000.000,-}$ (**Dua puluh lima juta rupiah**) = **Rp. 10.000.000,- (Sepuluh juta rupiah)**, dibayarkan setelah kontrak penelitian ini ditandatangani para pihak.
 - b. Pembayaran Tahap Kedua sebesar 40% dari total dana yaitu $40\% \times \text{Rp. 25.000.000,-}$ (**Dua puluh lima juta rupiah**) = **Rp. 10.000.000,- (Sepuluh juta rupiah)**, dibayarkan setelah **PIHAK KEDUA** menyerahkan *hard* dan *soft copy* berkas berikut:
 1. Draft Artikel *Review*
 2. Bukti Artikel *Review* berstatus *submit*.Seluruh berkas yang dibutuhkan untuk pencairan tahap kedua tersebut harus diserahkan ke DRHPM Universitas Padjadjaran paling lambat tanggal **30 April 2026**.
 - c. Pembayaran Tahap Ketiga sebesar 20% dari total dana yaitu $20\% \times \text{Rp. 25.000.000,-}$ (**Dua puluh lima juta rupiah**) = **Rp. 5.000.000,- (Lima juta rupiah)** dibayarkan setelah **PIHAK KEDUA** menyerahkan *Hard* dan *Soft copy* Bukti Artikel *Review* berstatus *under review*.
Seluruh berkas yang dibutuhkan untuk pencairan tahap ketiga tersebut harus diserahkan ke DRHPM Universitas Padjadjaran paling lambat tanggal **31 Juli 2026**.
- (3) **PIHAK KEDUA** bertanggungjawab mutlak dalam penggunaan dana seperti yang tercantum pada ayat (1) sesuai dengan proposal kegiatan yang telah disetujui dan berkewajiban untuk menyimpan semua bukti-bukti pengeluaran sesuai dengan jumlah dana yang diberikan oleh **PIHAK PERTAMA**.

PASAL 4 MEKANISME PEMBAYARAN

- (1) Dana Hibah Penulisan Artikel *Review Batch 2 Equity - WCU Unpad T.A. 2025* sebagaimana dimaksud Pasal 3 ayat (1) dibayarkan oleh **PIHAK PERTAMA** kepada **PIHAK KEDUA** melalui rekening *Virtual Account* Bank **PIHAK KEDUA**.
- (2) **PIHAK PERTAMA** tidak bertanggung jawab atas keterlambatan dan/atau tidak terbayarnya sejumlah dana sebagaimana dimaksud dalam pasal 3 ayat (1) yang disebabkan karena kesalahan **PIHAK KEDUA**.

PASAL 5 OUTPUT RISET

Output riset yang disepakati dalam perjanjian ini adalah satu (1) artikel review di jurnal Internasional terindeks minimal Scopus Q2 dengan status *under review* pada tahun pertama

dan diharuskan mencantumkan nomor kontrak; Hibah EQUITY-WCU Kemdiktisaintek; didanai oleh LPDP pada bagian *acknowledgement* sebagai pemberi hibah.

PASAL 6 HAK DAN KEWAJIBAN

- (1) Hak dan Kewajiban **PIHAK PERTAMA**:
 - a. **PIHAK PERTAMA** berhak menerima luaran/*output* dari **PIHAK KEDUA** sebagaimana dimaksud pada Pasal 5 penelitian yang telah sepakati dalam perjanjian ini
 - b. **PIHAK PERTAMA** berhak mengawasi, memonitor dan mengevaluasi kegiatan riset yang dilaksanakan oleh **PIHAK KEDUA**;
 - c. **PIHAK PERTAMA** berkewajiban menyediakan dana riset kepada **PIHAK KEDUA**, berdasarkan hasil penilaian proposal teknis dan penilaian pendanaan;
 - d. **PIHAK PERTAMA** menyediakan dan memberi informasi terkait kegiatan riset yang diperlukan oleh **PIHAK KEDUA**.
- (2) Hak dan Kewajiban **PIHAK KEDUA**:
 - a. **PIHAK KEDUA** berhak menerima dana riset dari **PIHAK PERTAMA** sesuai dengan hasil penilaian proposal, untuk dibiayai melalui Anggaran Universitas Padjadjaran PTN-BH;
 - b. **PIHAK KEDUA** berkewajiban:
 - 1) Mengunggah *softcopy* melalui akun staffs.unpad.ac.id berupa *Draft* Luaran Artikel *Review* dan Bukti Artikel *Review* berstatus *submit* kepada **PIHAK PERTAMA**, paling lambat tanggal **30 April 2026**.
 - 2) Mengunggah *softcopy* melalui akun staffs.unpad.ac.id berupa Bukti Artikel *Review* berstatus *under review*, kepada **PIHAK PERTAMA**, paling lambat tanggal **31 Juli 2026**;
 - 3) Bertanggung jawab terhadap penggunaan dana riset dalam bentuk laporan keuangan yang telah diterima dari **PIHAK PERTAMA** sesuai dengan Peraturan perundangan yang berlaku;
 - 4) Menyediakan dan memberi informasi terkait kegiatan riset yang diperlukan oleh **PIHAK PERTAMA**.

PASAL 7 MONITORING DAN EVALUASI

PIHAK PERTAMA dapat melakukan kegiatan *monitoring* dan evaluasi terhadap kemajuan pelaksanaan Hibah Penulisan Artikel *Review* Batch 2 Equity - WCU Unpad Tahun 2025.

PASAL 8 PERUBAHAN KETUA

- (1) Perubahan terhadap susunan Tim riset Hibah Penulisan Artikel *Review* Batch 2 Equity - WCU Unpad T.A. 2025 dapat dibenarkan apabila telah mendapat persetujuan tertulis dari **PIHAK PERTAMA**.
- (2) Apabila **PIHAK KEDUA** selaku Ketua Peneliti sebagaimana dimaksud pada Pasal 1 tidak dapat melaksanakan kegiatan Hibah Penulisan Artikel *Review* Batch 2 Equity - WCU Unpad T.A. 2025, maka **PIHAK KEDUA** wajib mengusulkan pengganti Ketua Peneliti kepada **PIHAK PERTAMA**.
- (3) Apabila **PIHAK KEDUA** tidak dapat melaksanakan tugas dan tidak ada pengganti Ketua sebagaimana dimaksud dalam ayat 2 maka **PIHAK KEDUA** harus mengembalikan dana kepada **PIHAK PERTAMA** yang selanjutnya disetor ke Kas Universitas Padjadjaran.

PASAL 9 JANGKA WAKTU

Perjanjian Pelaksanaan Hibah Penulisan Artikel *Review* Batch 2 Equity - WCU Unpad T.A. 2025 ini berlaku efektif sejak tanggal **22 Desember 2025** dan berakhir sampai *output* sebagaimana yang diatur pada Pasal 5 *under review*, selambat-lambatnya tanggal **31 Juli 2026**.

PASAL 10

SANKSI

- (1) Apabila sampai dengan batas waktu yang telah ditetapkan untuk melaksanakan Hibah Penulisan Artikel *Review Batch 2 Equity - WCU Unpad T.A. 2025* telah berakhir, **PIHAK KEDUA** belum menyelesaikan tugasnya dan atau terlambat mengirim laporan sebagaimana tercantum dalam Pasal 6 ayat (2), maka para periset dibawah koordinasi **PIHAK KEDUA** dapat dikenakan sanksi administratif.
- (2) Apabila **PIHAK KEDUA** tidak melaksanakan kewajiban memenuhi *output* riset sebagaimana yang tercantum dalam Pasal 5 dan Pasal 6 ayat (2) huruf b maka **PIHAK PERTAMA** akan memberikan sanksi berupa :
 - a. Meminta bantuan kepada instansi pemeriksa internal yang berwenang untuk melakukan audit terhadap kegiatan pemeriksaan langsung kepada **PIHAK KEDUA**;
 - b. Menghentikan pendanaan Riset berdasarkan Perjanjian ini;
 - c. **PIHAK KEDUA** tidak dapat mendaftar hibah riset dan hibah pengabdian pada masyarakat pada tahun berikutnya.

PASAL 11

DUPLIKASI

- (1) Apabila dikemudian hari judul Hibah Penulisan Artikel *Review Batch 2 Equity - WCU Unpad T.A. 2025* sebagaimana dimaksud pada Pasal 2 ditemukan adanya duplikasi dengan hibah lain dan/atau ditemukan adanya ketidak jujuran/itikad tidak baik yang tidak sesuai dengan kaidah ilmiah, maka Surat Perjanjian ini dinyatakan batal dan **PIHAK KEDUA** wajib mengembalikan seluruh dana yang telah diterima kepada **PIHAK PERTAMA**.
- (2) Bukti setor atas pengembalian dana sebagaimana dimaksud pada ayat (1) disimpan oleh kepada **PIHAK PERTAMA** selanjutnya akan diserahkan ke Universitas Padjadjaran.

PASAL 12

PAJAK DAN BIAYA LAINNYA

Ketentuan pengenaan pajak pertambahan nilai, pajak penghasilan dan/atau pajak-pajak lainnya dalam rangka pelaksanaan kegiatan penelitian ini wajib dilaksanakan oleh **PIHAK KEDUA** sesuai dengan ketentuan peraturan perundang-undangan dibidang perpajakan.

PASAL 13

KEPEMILIKAN

- (1) Hak atas kekayaan intelektual yang dihasilkan dari pelaksanaan Hibah Penulisan Artikel *Review Batch 2 Equity - WCU Unpad T.A. 2025* ini diatur dan dikelola sesuai dengan peraturan dan perundang-undangan yang berlaku.
- (2) Hasil Hibah Penulisan Artikel *Review Batch 2 Equity - WCU Unpad T.A. 2025* yang berupa peralatan dan/atau alat yang dibeli dari kegiatan ini adalah milik Negara yang dapat dihibahkan kepada institusi/lembaga/masyarakat melalui Surat Keterangan Hibah.

PASAL 14

KEADAAN KAHAR (*FORCE MAJEURE*)

- (1) Keadaan kahar (*force majeure*) adalah suatu keadaan yang terjadi di luar kehendak kedua belah pihak yang mempengaruhi pelaksanaan Kontrak ini sehingga **PEKERJAAN** yang telah ditentukan dalam Kontrak ini menjadi tidak dapat dipenuhi;
- (2) Hal-hal yang termasuk keadaan kahar (*force majeure*) sebagaimana tercantum pada ayat (1) Pasal ini adalah peperangan, kerusakan, revolusi, bencana alam (banjir, gempa bumi, badai, gunung meletus, tanah longsor, wabah penyakit dan angin topan), pemogokan, kebakaran dan gangguan industri lainnya, serta keadaan lainnya sesuai dengan Peraturan Perundangan yang berlaku;
- (3) Keterangan tentang kebenaran adanya keadaan kahar (*force majeure*) sebagaimana tercantum pada ayat (1) Pasal ini harus dibuat oleh instansi/pejabat yang berwenang;
- (4) Apabila terjadi keadaan kahar (*force majeure*) sebagaimana tercantum pada ayat (1) Pasal ini, maka **PIHAK KEDUA** wajib memberikan laporan tertulis paling lambat 14 (empat belas)

hari kalender berdasarkan laporan tertulis kepada **PIHAK PERTAMA**, dan atas laporan tertulis **PIHAK KEDUA** akan mengadakan penelitian oleh tim yang dibentuk oleh **PIHAK PERTAMA** dan instansi yang berwenang, yang kemudian berdasarkan Berita Acara hasil penelitian tersebut akan dilakukan penyelesaian lebih lanjut mengenai pelaksanaan Kontrak ini.

PASAL 15 **PENYELESAIAN PERSELISIHAN**

- (1) Apabila terjadi perselisihan antara **PIHAK PERTAMA** dan **PIHAK KEDUA** dalam pelaksanaan perjanjian ini akan dilakukan penyelesaian secara musyawarah dan mufakat, dan apabila tidak tercapai penyelesaian secara musyawarah dan mufakat maka penyelesaian dilakukan melalui proses hukum.
- (2) Hal-hal yang belum diatur dalam perjanjian ini diatur kemudian oleh kedua belah pihak.

PASAL 16 **PEMBATALAN DAN PENGAKHIRAN KONTRAK**

- (1) Kontrak ini dapat berakhir/atau diakhiri sebelum jangka waktu yang diperjanjikan jika:
 - a. Terjadi *force majeure*; dan
 - b. **PIHAK KEDUA** cidera janji atau tidak memenuhi kewajiban dan tanggungjawabnya sebagaimana tercantum dalam kontrak.
- (2) Dalam hal terjadi pengakhiran Kontrak ini sebagaimana dimaksud pada ayat (1) huruf a, di atas, maka kedua belah pihak akan berupaya melakukan hal sebagai berikut :
 - a. Merundingkan kemungkinan dilanjutkannya Kontrak ini; atau
 - b. Dalam hal kedua belah pihak sepakat untuk tidak melanjutkan Kontrak ini, maka akan dilakukan perhitungan secara proporsional, baik atas prestasi yang telah dilaksanakan oleh **PIHAK KEDUA** maupun atas kerugian yang mungkin ditimbulkan akibat pengakhiran Kontrak ini, yang besarnya akan disepakati oleh **PARA PIHAK**.
- (3) Pengakhiran Kontrak sebagaimana dimaksud ayat (1) huruf b Pasal ini dinyatakan secara tertulis dan sepihak oleh **PIHAK PERTAMA**.
- (4) Apabila terjadi pemutusan hubungan kerja oleh **PIHAK PERTAMA** akibat ayat (1) huruf b Pasal ini, maka **PIHAK KEDUA** tidak mendapat ganti rugi apa pun dan semua risiko menjadi tanggung jawab **PIHAK KEDUA**.
- (5) Apabila pelaksanaan kegiatan ini tidak selesai dalam jangka waktu yang telah ditetapkan, maka **PIHAK KEDUA** wajib mengembalikan sisa dana ke Kas Negara/Kas Universitas Padjadjaran dengan melaporkan kepada **PIHAK PERTAMA**.

PASAL 17 **AMANDEMEN**

Perubahan isi Kontrak ini dapat dilakukan sesuai kesepakatan **PARA PIHAK**, yang akan dituangkan dalam suatu Amandemen, yang merupakan bagian tidak terpisahkan dari Kontrak ini.

PASAL 18 **PENUTUP**

Surat Perjanjian Pelaksanaan Hibah Penulisan Artikel *Review* Batch 2 Equity - WCU Unpad T.A. 2025 ini dibuat rangkap 2 (dua) dan bermaterai cukup sesuai dengan ketentuan yang berlaku, dan biaya materai dibebankan kepada **PIHAK KEDUA**.



Prof. dr. Nur Atik, M.Kes., Ph.D.
NIP. 198110102008011019

PIHAK KEDUA



Dr. Restu Arisanti, S.Si., M.Si.
NIP. 198007192012122004

hari kalender berdasarkan laporan tertulis kepada **PIHAK PERTAMA**, dan atas laporan tertulis **PIHAK KEDUA** akan mengadakan penelitian oleh tim yang dibentuk oleh **PIHAK PERTAMA** dan instansi yang berwenang, yang kemudian berdasarkan Berita Acara hasil penelitian tersebut akan dilakukan penyelesaian lebih lanjut mengenai pelaksanaan Kontrak ini.

PASAL 15 **PENYELESAIAN PERSELISIHAN**

- (1) Apabila terjadi perselisihan antara **PIHAK PERTAMA** dan **PIHAK KEDUA** dalam pelaksanaan perjanjian ini akan dilakukan penyelesaian secara musyawarah dan mufakat, dan apabila tidak tercapai penyelesaian secara musyawarah dan mufakat maka penyelesaian dilakukan melalui proses hukum.
- (2) Hal-hal yang belum diatur dalam perjanjian ini diatur kemudian oleh kedua belah pihak.

PASAL 16 **PEMBATALAN DAN PENGAKHIRAN KONTRAK**

- (1) Kontrak ini dapat berakhir/atau diakhiri sebelum jangka waktu yang diperjanjikan jika:
 - a. Terjadi *force majeure*; dan
 - b. **PIHAK KEDUA** cidera janji atau tidak memenuhi kewajiban dan tanggungjawabnya sebagaimana tercantum dalam kontrak.
- (2) Dalam hal terjadi pengakhiran Kontrak ini sebagaimana dimaksud pada ayat (1) huruf a, di atas, maka kedua belah pihak akan berupaya melakukan hal sebagai berikut :
 - a. Merundingkan kemungkinan dilanjutkannya Kontrak ini; atau
 - b. Dalam hal kedua belah pihak sepakat untuk tidak melanjutkan Kontrak ini, maka akan dilakukan perhitungan secara proporsional, baik atas prestasi yang telah dilaksanakan oleh **PIHAK KEDUA** maupun atas kerugian yang mungkin ditimbulkan akibat pengakhiran Kontrak ini, yang besarnya akan disepakati oleh **PARA PIHAK**.
- (3) Pengakhiran Kontrak sebagaimana dimaksud ayat (1) huruf b Pasal ini dinyatakan secara tertulis dan sepihak oleh **PIHAK PERTAMA**.
- (4) Apabila terjadi pemutusan hubungan kerja oleh **PIHAK PERTAMA** akibat ayat (1) huruf b Pasal ini, maka **PIHAK KEDUA** tidak mendapat ganti rugi apa pun dan semua risiko menjadi tanggung jawab **PIHAK KEDUA**.
- (5) Apabila pelaksanaan kegiatan ini tidak selesai dalam jangka waktu yang telah ditetapkan, maka **PIHAK KEDUA** wajib mengembalikan sisa dana ke Kas Negara/Kas Universitas Padjadjaran dengan melaporkan kepada **PIHAK PERTAMA**.

PASAL 17 **AMANDEMEN**

Perubahan isi Kontrak ini dapat dilakukan sesuai kesepakatan **PARA PIHAK**, yang akan dituangkan dalam suatu Amandemen, yang merupakan bagian tidak terpisahkan dari Kontrak ini.

PASAL 18 **PENUTUP**

Surat Perjanjian Pelaksanaan Hibah Penulisan Artikel *Review* Batch 2 Equity - WCU Unpad T.A. 2025 ini dibuat rangkap 2 (dua) dan bermaterai cukup sesuai dengan ketentuan yang berlaku, dan biaya materai dibebankan kepada **PIHAK KEDUA**.



PIHAK PERTAMA
Prof. dr. Nur Atik, M.Kes., Ph.D.
NIP. 198110102008011019

PIHAK KEDUA

Dr. Restu Arisanti, S.Si., M.Si.
NIP. 198007192012122004



KEMENTERIAN PENDIDIKAN TINGGI, SAINS DAN TEKNOLOGI
UNIVERSITAS PADJADJARAN

Jalan Dipati Ukur No. 35 Bandung 40132
Jalan Ir. Soekarno Km. 21 Jatinangor, Sumedang 45363
Telepon (022) 84288888 Laman: www.unpad.ac.id, Email: humas@unpad.ac.id

**SURAT PERNYATAAN TANGGUNGJAWAB MUTLAK
HIBAH ARTIKEL *REVIEW* BATCH 2 EQUITY – WCU UNIVERSITAS PADJADJARAN
TAHUN ANGGARAN 2025**

Yang bertanda tangan di bawah ini:

Nama : **Dr. Restu Arisanti, S.Si., M.Si.**
NIP : 198007192012122004
Jabatan : Ketua Peneliti (Skema RKI-AR Batch 2)
Judul : Methodological Developments of Generalized Linear Mixed Models: A Systematic Review with Applications in Tuberculosis Epidemiology
Fakultas : Matematika dan Ilmu Pengetahuan Alam
No. kontrak : **5652/UN6.3.1/PT.00/2025, Tanggal 22 Desember 2025**
Jumlah Dana : **Rp. 25.000.000,- (Dua puluh lima juta rupiah)**

Menyatakan dengan sesungguhnya bahwa:

1. Bertanggungjawab mutlak dalam pembelanjaan dana Hibah Penulisan Artikel *Review* Batch 2 Equity - WCU Unpad T.A. 2025 dan berkewajiban untuk menyimpan semua bukti-bukti pengeluaran sesuai dengan jumlah dana yang diberikan oleh Universitas Padjadjaran melalui DRHPM Unpad;
2. Bertanggungjawab penuh atas data administrasi pelaksana penerima dana Hibah Penulisan Artikel *Review* Batch 2 Equity - WCU Unpad T.A. 2025.
3. Berkewajiban penuh mengunggah *softcopy* melalui akun staffs.unpad.ac.id berupa Hasil Artikel *Review* dan Bukti Artikel *Review* berstatus *under review*, paling lambat tanggal **30 April 2026** kepada **PIHAK PERTAMA**, sesuai ketentuan pada Buku Pedoman Hibah Penulisan Artikel *Review* pada Jurnal Internasional Bereputasi Universitas Padjadjaran;
4. Berkewajiban penuh mengunggah bukti *softcopy* Artikel *Review* Batch 2 berstatus *Accepted* melalui akun staffs.unpad.ac.id, paling lambat tanggal **31 Juli 2026** kepada **PIHAK PERTAMA**, sesuai ketentuan pada Buku Pedoman Hibah Penulisan Artikel *Review* pada Jurnal Internasional Bereputasi;
5. Berkewajiban untuk menindaklanjuti dan mengupayakan hasil luaran yang dijanjikan Hibah Penulisan Artikel *Review* Batch 2 Equity - WCU Unpad T.A. 2025 secara efektif dan efisien.

Jatinangor, 22 Desember 2025
Ketua Peneliti,



Dr. Restu Arisanti, S.Si., M.Si.
NIP. 198007192012122004

Generalized linear mixed models in epidemiological data analysis: A systematic review of methodological developments

Restu Arisanti^{a*}, Maizatul Akmar Ismail^b, Sri Winarni^a and Nita Cahyani^a

^aDepartment of Statistics, Faculty of Mathematics and Natural Sciences, Universitas Padjadjaran, Jatinangor 45363, Indonesia

^bDepartment of Information System, Faculty of Computer Science and Information Technology, Universiti Malaya, Malaysia

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Spatial dependence

ABSTRACT

Generalized linear mixed models (GLMMs) have become a fundamental statistical framework for analyzing hierarchical and correlated data in epidemiology and related quantitative fields. By combining non-Gaussian response distributions with random effects and structured dependence, GLMMs allow researchers to represent complex data structures arising from spatial, temporal, and multilevel processes. Over the past two decades, numerous methodological extensions have been introduced, including developments in spatial modeling, hierarchical structures, and computational inference. Despite this growth, these contributions are often presented as separate methodological advances, making it difficult to understand how they relate within the broader architecture of the GLMM framework. This study addresses this issue through a systematic review of methodological developments in GLMM-based epidemiological research. Thirty-nine studies published between 2001 and 2025 were examined to identify recurring modeling patterns and structural extensions of the GLMM framework. Based on this synthesis, a four-layer mathematical taxonomy is proposed that organizes GLMM methodology according to probabilistic specification, hierarchical structure, structured dependence, and inferential strategy. The results indicate that most innovations arise from structural modifications within these layers rather than from entirely new modeling paradigms, providing a clearer perspective on methodological diversity in the GLMM literature.

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1. Introduction

Generalized Linear Mixed Models (GLMMs) have become one of the most widely used statistical frameworks for analyzing complex and structured data. By extending generalized linear models through the inclusion of random effects, GLMMs enable researchers to model non-Gaussian responses while accounting for hierarchical dependence and correlation among observations. Since the seminal work of Breslow and Clayton (1993), the methodological landscape of GLMMs has expanded considerably, supported by advances in approximation techniques, variance component modeling, and computational inference. As a result, GLMMs are now routinely applied in many empirical settings where data arise from multilevel or correlated processes.

In epidemiological data analysis, the relevance of GLMMs becomes particularly apparent because health-related observations rarely satisfy the assumption of independence. Disease incidence and prevalence are typically recorded across multiple geographic regions, nested within administrative units, and often observed repeatedly over time. Moreover, epidemiological outcomes frequently exhibit features such as overdispersion, excess zeros, and spatial dependence. Early contributions in spatial and spatio-temporal disease modeling, including the studies of MacNab and Dean (2002) and Diggle et al. (2002), illustrated how random effects could represent latent spatial heterogeneity while maintaining a coherent likelihood-based inference framework. Subsequent empirical applications further demonstrated the flexibility of GLMMs for modeling geographically distributed health outcomes, including spatial prediction of mosquito-borne disease risk (Yoo, 2014) and large-scale disease mapping frameworks that accommodate spatial misalignment and regional heterogeneity (Hund et al.,

* Corresponding author

E-mail address r.arisanti@unpad.ac.id (R. Arisanti)

2012). Building on these foundations, subsequent work incorporated conditional autoregressive structures to capture spatial dependence (Escaramis et al., 2008), spline-based smoothing techniques to represent temporal dynamics (Torabi & Rosychuk, 2011), spatio-temporal disease risk modeling in epidemiological surveillance (Lowe et al., 2013), and reduced-rank spatial representations designed to mitigate spatial confounding (Hughes & Haran, 2013). Additional studies extended these approaches to broader epidemiological contexts, including spatio-temporal mortality analysis in demographic surveillance systems (Escaramis et al., 2011) and regional modeling of dengue transmission patterns (Abd Naeem et al., 2022).

Parallel to developments in model specification, considerable progress has also been made in the statistical inference of GLMMs. Early estimation approaches, such as penalized quasi-likelihood and empirical Bayes procedures (MacNab & Lin, 2009), provided practical tools for fitting hierarchical models in disease mapping applications. Later methodological advances introduced Bayesian frameworks based on latent Gaussian models, particularly through the integrated nested Laplace approximation (INLA) methodology (Rue et al., 2009), which allowed more efficient inference for spatial and spatio-temporal models. The stochastic partial differential equation (SPDE) formulation proposed by Lindgren, Rue, and Lindström (2011) further strengthened this framework by establishing a link between Gaussian random fields and Gaussian Markov random fields, thereby allowing spatial dependence structures to be represented through sparse precision matrices. Together, these developments substantially improved the scalability and practical implementation of spatial GLMMs in large epidemiological datasets.

Despite these advances, several methodological challenges remain central to GLMM-based epidemiological modeling. Issues related to parameter identifiability, spatial confounding, and variance decomposition continue to attract significant attention in the literature. For example, Goicoa et al. (2018) demonstrated that identifiability constraints in spatio-temporal models may influence the interpretation of spatial random effects. Similarly, Hughes and Haran (2013) highlighted how interactions between fixed and random effects can lead to spatial confounding, complicating the interpretation of regression parameters. More recent studies have also addressed challenges associated with penalized likelihood under covariate rank deficiency (Krause et al., 2022) as well as Bayesian regularization techniques designed to stabilize spatial models (Hefley et al., 2017) alongside spatio-temporal risk modeling for infectious diseases (Lubinda et al., 2021). These contributions illustrate that ongoing methodological innovation in GLMMs often emerges from attempts to address deeper structural issues in hierarchical data analysis.

Although these methodological contributions share a common statistical foundation, they frequently modify different components of the GLMM framework. For instance, small area estimation models derived from the Fay–Herriot approach (Yilema, 2025), copula-based mixed models for diagnostic test meta-analysis (Nikolouloupoulos, 2019), hierarchical models for zero-inflated count outcomes (Tawiah et al., 2020), and Bayesian mixed-effects regression models for group testing data (McMahan et al., 2025) represent distinct directions of methodological development. Earlier methodological contributions also explored correlated random-effects structures in diagnostic test evaluation (Chu et al., 2009) and Bayesian modeling strategies for epidemiological screening processes (Armero et al., 2008). However, the existing literature rarely organizes these contributions within a unified analytical perspective that clarifies how such extensions relate to the broader structure of GLMM methodology.

Despite the growing body of methodological contributions, the literature lacks a systematic synthesis that organizes these developments from a unified modelling perspective. Existing studies typically focus on specific model extensions, such as spatial models, Bayesian estimation, or hierarchical regression, without examining how these methodological advances relate within the broader GLMM framework. As a result, the structural evolution of GLMM methodology in epidemiological data analysis remains fragmented and difficult to interpret from a unified analytical standpoint.

Formally, a generalized linear mixed model may be expressed as

$$Y_i | \mathbf{u} \sim F(\mu_i, \boldsymbol{\theta}), g(\mu_i) = X_i \boldsymbol{\beta} + Z_i \mathbf{u}, \quad (1)$$

where F denotes a distribution from the exponential family, $g(\cdot)$ represents the link function, and $\mathbf{u} \sim N(0, \Sigma_u)$ corresponds to latent random effects. In spatial and spatio-temporal extensions, the covariance structure of the random effects is frequently represented through structured precision matrices that encode spatial dependence using conditional autoregressive or Gaussian Markov random field formulations. Inference for such models typically requires evaluation of the marginal likelihood

$$L(\boldsymbol{\beta}, \boldsymbol{\theta}) = \int \prod_i f(Y_i | \mu_i) f(\mathbf{u} | \boldsymbol{\theta}) d\mathbf{u}, \quad (2)$$

which involves a high-dimensional integration problem that motivates approximation strategies such as Laplace approximation, numerical quadrature, or Bayesian latent Gaussian inference.

This representation highlights that GLMM methodology can be viewed as a layered modelling architecture involving several interconnected components, including the probabilistic data distribution, the hierarchical random-effect structure, the specification of dependence patterns, and the inferential framework used for parameter estimation. Interactions among these components often generate practical challenges related to identifiability, computational scalability, and interpretation of model parameters. Nevertheless, methodological developments in the literature are typically discussed separately rather than within a unified conceptual structure.

To address this gap, the present study conducts a systematic review of methodological developments in generalized linear mixed models applied to epidemiological data analysis between 2001 and 2025. The objectives of this study are threefold. First, to synthesize the evolution of GLMM-based modelling approaches used in epidemiological research. Second, to organize these developments into a structured methodological taxonomy that clarifies how different modelling extensions modify the GLMM framework. Third, to identify recurring methodological challenges and emerging directions for future research in epidemiological data modelling.

By organizing the literature within a unified analytical perspective, this review aims to provide a clearer understanding of how GLMM methodology has evolved to address increasingly complex epidemiological data structures. The proposed taxonomy offers a structured overview of existing modelling approaches and highlights potential opportunities for future methodological development in hierarchical and spatial epidemiological analysis.

2. Research Methodology

2.1. Database and Search Strategy

The literature search was conducted using the Scopus database. Scopus was selected because of its extensive coverage of peer-reviewed journals in statistics, applied mathematics, and quantitative epidemiology. Many foundational and methodological contributions to generalized linear mixed models (GLMMs) have appeared in journals indexed by Scopus, including *Journal of the American Statistical Association*, *Journal of the Royal Statistical Society*, *Biometrics*, and *Spatial Statistics*. For a review centered on methodological structure rather than applied case reporting, this scope of coverage was considered appropriate.

The search period extended from January 2001 to December 2025. The lower bound was chosen to capture developments that followed early spatio-temporal GLMM formulations in disease mapping (MacNab & Dean, 2002; Diggle *et al.*, 2002), which signaled an important shift toward structured hierarchical and spatial modeling in epidemiology. Subsequent empirical studies further demonstrated the applicability of GLMMs to geographically structured epidemiological data, including spatial risk prediction for vector-borne diseases (Yoo, 2014) and large-scale disease mapping with spatial misalignment (Hund *et al.*, 2012). Extending the time frame through 2025 made it possible to observe more recent advances in Bayesian inference, spatial precision modeling, and regularized estimation.

Search terms were constructed to balance specificity and breadth. Keywords such as “generalized linear mixed model”, “GLMM”, “hierarchical model”, “spatial”, “spatio-temporal”, “disease mapping”, and “small area estimation” were combined using Boolean operators. The search strategy was designed to identify studies that engaged with the structural components of GLMMs, including likelihood specification, random effects formulation, and dependence modeling, rather than studies that merely applied GLMMs as standard analytical tools.

Only journal articles written in English were retained. Conference proceedings, book chapters, and editorial materials were excluded to maintain methodological consistency and peer-reviewed quality. The initial search yielded 586 records. Subsequent stages of screening and eligibility assessment are described in Sections 2.2 and 2.3.

2.2 Eligibility Criteria and Relevance Filtering

Because generalized linear mixed models are widely used across applied research, a relevance filtering stage was necessary to distinguish structural methodological contributions from routine applications. The objective of this review was not to summarize epidemiological findings, but to synthesize developments that modified or clarified the mathematical structure of GLMMs. Studies were considered eligible if they satisfied at least one of the following criteria. First, the study proposed a methodological extension of GLMMs, such as modifications to the likelihood, link function, or distributional assumptions. Second, the study introduced or refined a hierarchical random effects formulation, including multilevel, area-level, or variance component structures. Third, the study incorporated structured dependence, such as spatial, temporal, or spatio-temporal correlation modeled through conditional autoregressive, Gaussian Markov random field, spline-based, or related frameworks. Fourth, the study developed or evaluated inferential strategies for GLMMs, including approximation methods, Bayesian latent Gaussian inference, or regularization under identifiability constraints.

Studies were excluded if GLMMs were used solely as standard analytical tools without substantive structural development. Applied studies focused exclusively on disease prediction or epidemiological interpretation were not retained unless they contributed new modeling insight. Similarly, studies that mentioned mixed models without explicit specification of the probabilistic or hierarchical structure were excluded.

This filtering stage was designed to ensure conceptual coherence across the final sample. In methodological reviews of spatial and hierarchical models, the unit of analysis is the modeling framework itself rather than the applied outcome domain, a principle emphasized in discussions of spatial confounding and identifiability (Hughes & Haran, 2013; Goicoa *et al.*, 2018). The same perspective guided the present selection process. After applying the mixed-model relevance filter, 256 studies remained for further screening.

2.3 Study Selection Process

After the initial relevance filtering, the remaining records were screened based on their titles and abstracts. At this stage, we carefully examined whether the use of generalized linear mixed models (GLMMs) was central to the study's design rather than simply a routine analytical choice. Papers were excluded when the mixed-model component appeared marginal, insufficiently described, or not meaningfully connected to the study's methodological framework.

From the 256 records that passed the relevance filtering stage, 169 were excluded during title and abstract screening. In many of these cases, GLMM was mentioned only briefly, without clear specification of the random-effects structure or model assumptions. The remaining 87 articles were considered potentially eligible and were therefore assessed in full text.

The full-text review involved a more detailed examination of each study's modeling approach. We looked for explicit model specification, including distributional assumptions, link functions, and the treatment of hierarchical or dependent data structures. Studies were excluded at this stage when the methodological contribution was limited, repetitive, or insufficiently developed in relation to the GLMM framework. Following this evaluation, 48 additional articles were excluded.

In total, 39 studies met all predefined inclusion criteria and were included in the qualitative synthesis. The full list of included studies and their bibliographic information is provided in Table A1 (Appendix A). This stepwise selection process allowed us to focus specifically on studies that provided clear and substantive contributions to the structure and application of GLMMs. The overall selection flow is summarized in Fig. 1. The progressive reduction from 586 initial records to 39 included studies reflects a deliberate and structured selection process aimed at isolating methodologically substantive GLMM contributions. This refinement ensures that the final corpus represents studies with clearly articulated model structures and meaningful engagement with hierarchical or dependence modeling.

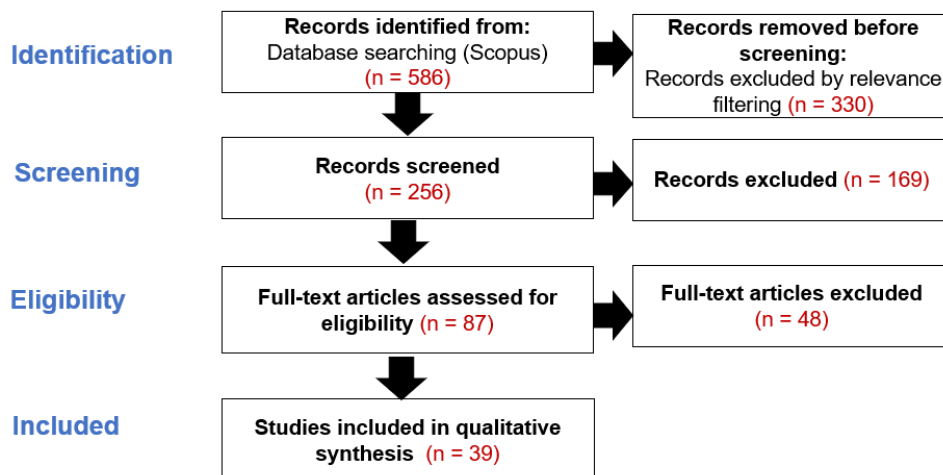


Fig. 1. PRISMA 2020 flow diagram of the study selection process

2.4 Data Extraction Framework

Generalized linear mixed models (GLMMs) extend the generalized linear modeling framework by incorporating random effects to accommodate hierarchical, clustered, or otherwise dependent data structures (Breslow and Clayton, 1993; McCulloch *et al.*, 2008). Over the past decades, methodological developments have expanded both the inferential strategies and structural formulations of GLMMs, including likelihood-based approximations, quasi-likelihood approaches, and Bayesian estimation paradigms (Lee *et al.*, 2006; Jiang, 2007). In light of this diversity, a structured data extraction framework was developed to ensure systematic and comparable synthesis across the included studies.

The objective of the extraction process was not to summarize substantive findings, but to document how GLMMs were formulated and implemented from a modeling perspective. For each of the 39 eligible studies, information was extracted using a predefined matrix designed to capture core inferential and structural components. The following elements were recorded:

- The estimation paradigm employed, including maximum likelihood (ML), restricted maximum likelihood (REML), Laplace approximation, adaptive Gauss–Hermite quadrature, penalized quasi-likelihood (PQL), or Bayesian approaches such as Markov chain Monte Carlo (MCMC);
- The assumed distributional family of the response variable, consistent with the exponential family framework underlying GLMM theory (McCulloch *et al.*, 2008);
- The link function connecting the linear predictor to the conditional mean structure;
- The specification of random effects, including random intercept models, random slope models, nested or crossed structures, and spatio-temporal extensions;
- The type of dependence structure addressed, such as clustered, longitudinal, hierarchical, spatial, or spatio-temporal correlation;
- The presence of structured modeling extensions, including zero-inflation, robustness adjustments, penalization strategies, copula-based dependence modeling, or other non-standard formulations;
- The computational environment or software implementation, when explicitly reported.

This extraction strategy reflects the dual nature of GLMMs as both inferential and structural modeling tools. By documenting distributional assumptions, random-effects structures, and estimation techniques in parallel, the framework enables meaningful comparison across heterogeneous application domains while remaining grounded in the theoretical foundations of mixed modeling (Jiang, 2007).

All extracted information was reviewed iteratively to maintain internal consistency. When model specifications were ambiguously reported, classification relied strictly on explicit methodological descriptions provided in the original articles. This approach ensured transparency and reproducibility, and it forms the basis for the classification strategy described in the following section. The extracted variables and their operational definitions are summarized in Table 1.

Table 1

Extracted Structural and Inferential Components of GLMM Formulation

Component	Extracted Variable	Operational Definition
Estimation Paradigm	Estimation Method	Inferential approach used for parameter estimation (e.g., ML, REML, PQL, Laplace approximation, adaptive quadrature, Bayesian MCMC)
Distributional Structure	Response Distribution	Assumed distributional family of the response variable (e.g., Binomial, Poisson, Negative Binomial, Gaussian)
Link Specification	Link Function	Function connecting the conditional mean of the response to the linear predictor (e.g., logit, log, identity, probit)
Random Effects	Random-Effects Structure	Specification of random intercepts, random slopes, nested, crossed, or spatio-temporal effects
Dependence Modeling	Correlation Structure	Type of hierarchical or dependence structure addressed (clustered, longitudinal, spatial, spatio-temporal)
Structural Extensions	Model Extensions	Presence of zero-inflation, copula-based dependence, robustness adjustments, penalization, or other structured modifications
Computational Implementation	Software/Algorithm	Reported computational tools or algorithms used for model fitting

2.5 Classification Strategy

The studies included in this review exhibit substantial methodological diversity in both the formulation and implementation of generalized linear mixed models (GLMMs). While these contributions share a common statistical foundation, they differ in how specific components of the model are specified, extended, or interpreted. To systematically organize these developments and facilitate comparative analysis, this study introduces a unified four-layer mathematical taxonomy of GLMM methodology. Rather than classifying studies by application domain, the proposed framework focuses on the internal structural components of the GLMM formulation.

This structural perspective allows methodological developments to be examined in terms of how they modify different elements of the model architecture. In many cases, innovations in GLMM methodology do not replace the entire modeling framework but instead extend or refine specific layers of the model. For instance, some studies introduce alternative distributional assumptions for the response variable, while others modify the random-effects structure, incorporate spatial dependence, or propose new inferential strategies. By organizing these developments according to their position within the

model structure, the taxonomy provides a coherent way to compare methodological contributions that might otherwise appear disconnected.

The proposed taxonomy decomposes GLMM formulation into four conceptually distinct but hierarchically related layers. Each layer corresponds to a particular mathematical dimension of the model, and modifications introduced at higher layers typically preserve the assumptions established in the lower layers. This hierarchical organization reflects the modular nature of GLMMs, where model components can be extended or refined without altering the entire probabilistic framework. The resulting classification therefore highlights how methodological innovation occurs within specific components of the GLMM architecture while maintaining the overall structure of the model.

Layer 1: Probabilistic Data-Generating Layer

At the most fundamental level, a GLMM begins with the specification of a conditional distribution for the observed response variable. Observations are assumed to arise from a distribution belonging to the exponential family, with parameters governing the conditional mean and dispersion. Common choices include Poisson, Negative Binomial, Binomial, and Gaussian distributions.

The choice of distribution at this layer determines the basic stochastic properties of the outcome variable and establishes the primary mean–variance relationship. In many epidemiological applications, particularly those involving count data or binary outcomes, deviations from the canonical variance structure may arise due to unobserved heterogeneity or clustering. Consequently, distinguishing between distributional dispersion and variance introduced by higher-level random effects becomes an important modeling consideration. The probabilistic layer therefore forms the foundation upon which all subsequent hierarchical and dependence structures are constructed.

Layer 2: Hierarchical Linear Predictor and Random Effects

Conditional on the distributional specification, the systematic component of the model is defined through a link function and a hierarchical linear predictor. At this level, fixed effects represent population-level associations between covariates and the outcome variable, while random effects capture latent heterogeneity across clusters, groups, or spatial units.

Typical structures at this layer include random intercept models, random slope models, nested designs, and crossed random effects. Conceptually, this layer separates within-unit variability from between-unit variability. The classical variance decomposition illustrates this distinction, in which total variance can be viewed as the sum of the expected conditional variance and the variance of the conditional expectations.

In practice, however, the distinction between dispersion and random-effect variance may become difficult to identify, particularly in datasets with small cluster sizes or sparse observations. As a result, methodological developments at this layer often focus on improving identifiability, stabilizing variance component estimation, or refining hierarchical model structures.

Layer 3: Structured Dependence Modeling

Beyond unstructured random effects, many GLMM extensions incorporate explicit dependence structures to capture spatial, temporal, or spatio-temporal correlation. At this stage, the covariance structure of the latent effects is no longer assumed independent but is modeled using structured correlation or precision matrices.

Representative approaches at this layer include spatial autoregressive models, conditional autoregressive (CAR) specifications, Gaussian process models, and spline-based temporal smoothers. These approaches allow the model to account for structured correlation patterns that arise in geographically or temporally indexed data.

Introducing structured dependence substantially enriches the modeling framework but also introduces additional inferential challenges. In particular, spatial confounding may occur when fixed effects and spatial random effects capture similar patterns of variation. Moreover, high-dimensional covariance structures may lead to numerical instability or computational complexity. Methodological contributions at this layer therefore often focus on improving identifiability, reducing dimensionality, or developing computationally efficient representations of dependence structures.

Layer 4: Inferential and Computational Strategy

The final layer concerns statistical inference and computational implementation. Because GLMMs involve integration over latent random effects, the marginal likelihood typically does not admit a closed-form solution. As a result, estimation procedures rely on approximation or simulation-based methods.

Frequentist approaches commonly employ techniques such as Laplace approximation, adaptive Gauss–Hermite quadrature, penalized quasi-likelihood, or restricted maximum likelihood estimation. Bayesian approaches, on the other hand, rely on computational strategies such as Markov chain Monte Carlo (MCMC) sampling or integrated nested Laplace approximation (INLA). At this layer, methodological considerations extend beyond theoretical formulation to include approximation accuracy, computational scalability, convergence behavior, and numerical stability. In recent years, advances in

computational statistics have substantially expanded the practical applicability of GLMMs to large and complex epidemiological datasets. The four-layer classification framework developed in this study is summarized in Table 2.

Table 2
Four-layer taxonomy of generalized linear mixed model methodology

Layer	Model Component	Role in GLMM	Typical Methodological Extensions
Layer 1	Probabilistic distribution	Defines the stochastic behavior of the response variable	Poisson, Binomial, Negative Binomial, Gaussian
Layer 2	Hierarchical linear predictor	Captures multilevel heterogeneity through random effects	Random intercepts, random slopes, nested or crossed effects
Layer 3	Structured dependence	Models spatial, temporal, or spatio-temporal correlation	CAR models, Gaussian processes, spline-based smoothers
Layer 4	Inferential and computational strategy	Enables estimation of model parameters	Laplace approximation, adaptive quadrature, MCMC, INLA

3. Results

3.1 Descriptive Overview of Included Studies

The 39 studies retained for qualitative synthesis span the period from 2001 to 2025. To facilitate interpretation of long-term publication patterns, the studies were grouped into five-year intervals. The distribution of publications across these periods is presented in Fig. 2.

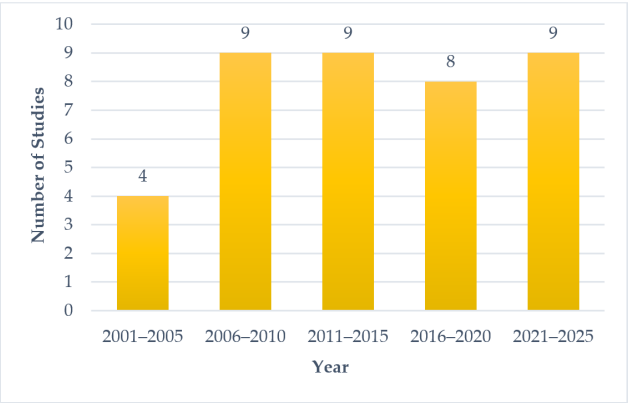


Fig. 2. Temporal Distribution of Included Studies (2001–2025)

During the period 2001–2005, four studies were identified, reflecting the relatively limited use of generalized linear mixed models (GLMMs) in epidemiological research at that time. The subsequent period (2006–2010) shows a noticeable increase, with nine studies published. A similarly elevated level was observed in 2011–2015, again totaling nine studies, suggesting a phase of methodological consolidation.

Between 2016 and 2020, the number of publications decreased slightly to eight studies, although the overall level remained comparable to earlier peak periods. The most recent interval, 2021–2025, again recorded nine studies, indicating sustained methodological interest in GLMM-based approaches within epidemiological and related quantitative fields.

Earlier publications in the corpus tended to focus on relatively standard mixed-model formulations with simple hierarchical structures. More recent studies increasingly incorporate structured dependence components, including spatial or spatio-temporal random effects, together with more advanced estimation strategies. This gradual shift likely reflects broader developments in computational statistics and the growing demand for models capable of accommodating hierarchical, spatial, and high-dimensional data structures.

Rather than indicating a sudden methodological shift, the observed trajectory suggests a gradual expansion of the GLMM framework. Methodological extensions such as spatial random effects, spatio-temporal structures, and Bayesian estimation approaches appear progressively over time, building upon earlier hierarchical modeling foundations.

Although the present review is organized primarily around model structure rather than application domain, the thematic context of the included studies remains informative. A substantial portion of the literature originates from epidemiology and public health, where clustered observations and spatially correlated data are common. Additional contributions appear in environmental modeling, econometrics, and related quantitative disciplines.

Despite this diversity of application areas, the underlying statistical architecture remains largely consistent. Across domains, researchers rely on exponential-family distributions combined with random-effects formulations to capture latent

heterogeneity. Differences across studies tend to arise not from the basic model form, but from how dependence structures and inferential strategies are implemented.

To further explore the thematic organization of the literature, a keyword co-occurrence network was constructed using the author keywords reported in the selected studies.

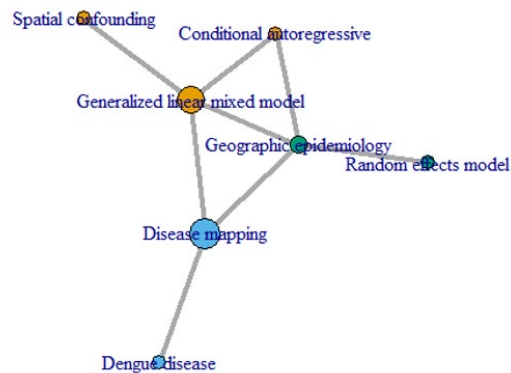


Fig. 3. Keyword Co-occurrence Network of GLMM Applications in Epidemiology

Fig. 3 illustrates the keyword co-occurrence network derived from the reviewed corpus. The network reveals a central cluster organized around generalized linear mixed models, reflecting the role of GLMMs as the methodological backbone of the literature. Closely connected to this core are spatial modeling concepts, including conditional autoregressive models and spatial confounding, which appear as prominent methodological extensions within the network.

From an applied perspective, disease mapping and geographic epidemiology emerge as dominant research domains in which GLMM-based approaches are frequently employed. Specific disease contexts, such as dengue, appear as smaller thematic clusters linked to the broader methodological framework. The overall topology of the network suggests that GLMM research in epidemiology is strongly oriented toward spatial hierarchical modeling, with many methodological developments occurring within spatial and Bayesian modeling paradigms.

A preliminary examination of the corpus also indicates that Poisson, Binomial, and Negative Binomial specifications are the most frequently encountered distributional choices. Random intercept models are widely used in clustered or multilevel data settings, whereas more elaborate formulations such as random slopes or structured spatial components appear in a subset of studies. Estimation strategies range from likelihood-based approximations to Bayesian computational approaches, reflecting the methodological diversity that motivates the layered synthesis presented in the following sections.

Overall, the evidence from the reviewed studies suggests a pattern of both stability and gradual diversification in the GLMM literature. Although the core formulation of the model has remained largely unchanged, researchers have progressively expanded the framework by introducing richer dependence structures and more sophisticated estimation strategies. These observations provide a descriptive foundation for the methodological synthesis presented in the following sections. Building on this descriptive overview, the next section examines the methodological characteristics of the reviewed studies, focusing on how generalized linear mixed models are specified, extended, and estimated across the literature.

3.2 Methodological Landscape of GLMM Applications

This section examines the methodological characteristics of the 39 studies included in the final corpus. Rather than summarizing substantive findings, the analysis focuses on how generalized linear mixed models (GLMMs) are formulated and implemented across the reviewed literature. Particular attention is given to distributional assumptions, hierarchical model structures, dependence modeling, and estimation strategies, as these elements collectively shape the statistical behavior of the models.

Distributional Assumptions and Link Functions

Most models in the reviewed corpus are formulated within the exponential-family framework that underlies generalized linear modeling. Count-based outcomes dominate the literature, and accordingly Poisson and Negative Binomial distributions appear most frequently. These specifications are commonly used when modeling disease incidence, event counts, or aggregated epidemiological outcomes.

In several studies, evidence of overdispersion prompted a shift from standard Poisson models to Negative Binomial formulations or to hierarchical models incorporating additional random effects. A smaller number of studies adopted zero-inflated or hurdle models in order to address excess zeros in observed counts. Across these settings, link functions generally

align with the assumed distribution, with log and logit links being the most common. These distributional choices form the probabilistic foundation of GLMMs and determine how variability is initially represented within the modeling framework.

Hierarchical Structures and Random Effects

Beyond the choice of response distribution, much of the methodological variation in the literature arises from the specification of random effects. Random intercept models are by far the most frequently used formulation, providing a straightforward mechanism for accounting for clustered observations across geographic regions, institutions, or population groups.

Random slope models appear less frequently but are used in studies where the effect of explanatory variables is expected to vary across clusters. In such cases, allowing regression coefficients to differ across observational units offers a more flexible representation of heterogeneity.

More complex hierarchical configurations also appear in a subset of studies. These include nested multilevel models, crossed random effects, and area-level formulations designed for spatially aggregated data. Interestingly, the reviewed literature suggests that hierarchical complexity is often introduced selectively, typically when the research question explicitly involves multilevel variation. In many cases, relatively simple random-intercept structures remain sufficient for capturing the dominant sources of heterogeneity.

Structured Dependence Components

A distinct line of methodological development concerns the incorporation of structured dependence, particularly in spatial and temporal contexts. Unlike standard random-effects formulations, these models explicitly encode correlation patterns among observations. Spatial dependence is most commonly modeled through conditional autoregressive (CAR) specifications or related precision-based representations. Such models allow neighboring regions to share information, thereby capturing spatial autocorrelation that cannot be explained by observed covariates alone.

Temporal dependence is typically represented through autoregressive processes or random-walk structures, especially in longitudinal or panel data settings. In a smaller number of studies, spatial and temporal components are combined within spatio-temporal models. These formulations substantially increase model flexibility, but they also introduce additional computational and inferential challenges.

Estimation Strategies and Computational Approaches

The choice of estimation strategy is closely tied to the structural complexity of the model. For models with relatively simple hierarchical structures, likelihood-based approaches remain widely used. Techniques such as Laplace approximation and adaptive Gauss–Hermite quadrature are commonly applied to approximate the marginal likelihood in the presence of random effects.

As model structures become more elaborate particularly when spatial or spatio-temporal dependence is introduced researchers increasingly rely on Bayesian computational approaches. Methods based on Markov chain Monte Carlo (MCMC) sampling or integrated nested Laplace approximation (INLA) provide practical solutions for estimating models with high-dimensional latent structures. In this sense, estimation strategies do not evolve independently of model specification. Instead, computational choices tend to follow the structural demands imposed by hierarchical and dependence components of the model.

Viewed collectively, the reviewed studies suggest that the GLMM framework itself has remained conceptually stable while its practical implementation has diversified over time. Differences across studies arise largely from how researchers introduce dependence structures and adapt estimation strategies to increasingly complex data settings. These patterns motivate the structural perspective developed in the next section, where methodological developments are organized within a unified four-layer taxonomy of GLMMs. This pattern reflects the predominance of count-based epidemiological outcomes in applied GLMM research.

3.3. A Four-Layer Mathematical Taxonomy of Generalized Linear Mixed Models

To move beyond a purely descriptive synthesis of the literature, this study introduces a unified four-layer mathematical taxonomy that organizes methodological developments in generalized linear mixed models according to the structural components of model construction. Within this framework, GLMM formulations identified in the reviewed corpus are interpreted through distinct conceptual layers, allowing methodological variations to be examined according to their structural role within the model.

When the 39 reviewed studies are examined through this layered perspective, patterns that initially appear scattered across the literature become more interpretable. Differences between studies can often be traced to distinct components of model construction rather than to entirely separate methodological approaches. Some contributions primarily refine the probabilistic assumptions of the response distribution, whereas others introduce additional hierarchical structures, structured dependence mechanisms, or alternative computational strategies. By distinguishing these layers of model formulation, the taxonomy

provides a clearer view of how GLMM methodology has gradually expanded across diverse application settings. The conceptual structure of the taxonomy is illustrated in Figure 4. By organizing GLMM formulations according to distinct layers of model construction, this framework provides a unified perspective for interpreting methodological developments that have previously appeared fragmented across the literature.

Layer 1: Likelihood Specification in the Examined Corpus

At the most fundamental level, GLMMs are defined by the probabilistic distribution assumed for the response variable. Mapping the reviewed studies onto this layer reveals a clear predominance of exponential-family distributions. In particular, Poisson and Negative Binomial specifications appear frequently in studies analyzing incidence counts, event frequencies, or aggregated epidemiological outcomes.

In several cases, evidence of overdispersion prompted researchers to move beyond the standard Poisson formulation and adopt Negative Binomial models. A smaller number of studies introduced zero-inflated or related extensions to accommodate excess zeros in the observed data. These adjustments typically arise from empirical characteristics of the data rather than from theoretical reformulations of the GLMM framework itself.

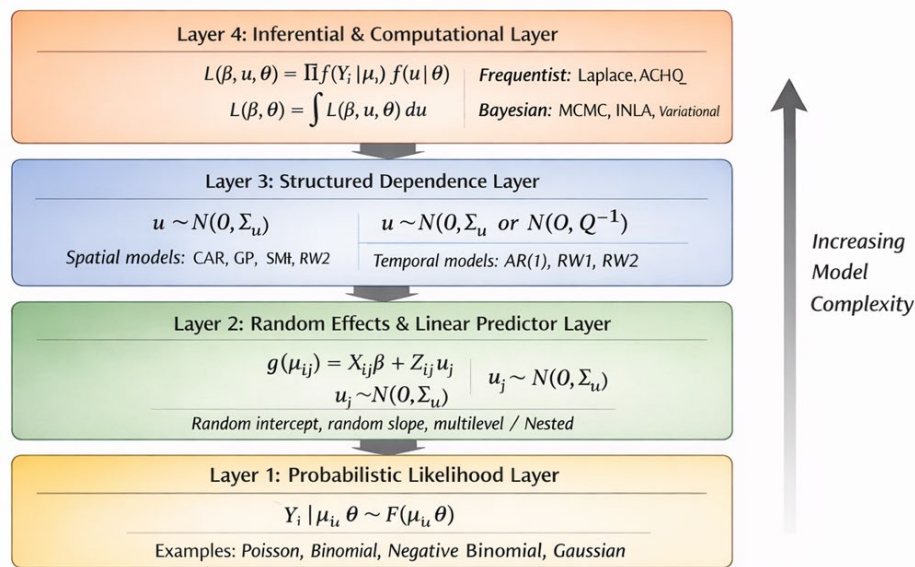


Fig. 4. A Unified Four-Layer Mathematical Taxonomy of Generalized Linear Mixed Models.

Consequently, variation at the probabilistic layer remains relatively limited compared with the higher layers of the taxonomy. Most methodological developments do not alter the fundamental distributional structure of the model, but instead build upon it through additional hierarchical or dependence components.

Layer 2: Hierarchical Structures and Random Effects

The second layer of the taxonomy concerns the hierarchical representation of latent heterogeneity through random effects. Across the reviewed studies, random intercept models constitute the most widely used specification. This formulation provides a practical way to account for clustering within groups such as geographic regions, institutions, or population subgroups.

Random slope models appear less frequently but are employed in studies where the influence of explanatory variables is expected to vary across observational units. These models allow regression coefficients to differ across clusters, thereby capturing heterogeneity in covariate effects.

More elaborate hierarchical designs were also observed in a subset of studies, including nested multilevel structures and crossed random-effects configurations. Such extensions tend to appear in analyses involving large datasets or explicitly multilevel research designs. Despite these variations, the general pattern suggests that hierarchical complexity often remains moderate, with most studies relying on relatively parsimonious random-effects structures.

Layer 3: Structured Dependence

A more pronounced methodological divergence emerges at the third layer, where models incorporate structured forms of dependence. In contrast to unstructured random effects, this layer introduces explicit correlation patterns across spatial or temporal dimensions.

Spatial dependence is most commonly represented using conditional autoregressive (CAR) models or related precision-matrix formulations that encode neighborhood relationships between spatial units. Temporal correlation is typically captured through autoregressive processes or random-walk structures, particularly in longitudinal or panel-data settings.

Although less frequently encountered, spatio-temporal models represent an important methodological direction, as they combine spatial and temporal components within a unified hierarchical framework. The incorporation of such dependence structures substantially increases model complexity and often requires more advanced estimation techniques. In many cases, these models combine structured correlation with traditional random effects, indicating that simple cluster-level heterogeneity alone may be insufficient to explain observed patterns in the data.

Layer 4: Inferential and Computational Strategies

The final layer of the taxonomy concerns the estimation and computational procedures used to fit GLMMs. Because marginal likelihoods involve integration over latent random effects, estimation typically relies on approximation or simulation-based techniques.

Likelihood-based approximations remain widely used in models with moderate complexity. Methods such as Laplace approximation and adaptive Gauss–Hermite quadrature are frequently applied to approximate the marginal likelihood. In models involving approximately Gaussian responses, restricted maximum likelihood (REML) is occasionally used.

When models incorporate spatial dependence or high-dimensional random-effect structures, Bayesian approaches appear more frequently. In these cases, estimation commonly relies on Markov chain Monte Carlo (MCMC) sampling or integrated nested Laplace approximation (INLA). These computational strategies provide greater flexibility for handling complex dependence structures and large latent variable spaces.

The reviewed studies suggest that the choice of estimation method is closely tied to the structural demands of the model. As additional hierarchical layers and dependence structures are introduced, computational considerations become increasingly central to the modeling process. In this sense, the inferential layer adapts to the complexity generated in the preceding layers, reinforcing the layered interpretation of GLMM methodology.

To further examine how methodological developments are distributed across the proposed taxonomy, the reviewed studies were mapped across time according to the four structural layers of GLMM construction. The resulting temporal pattern is summarized in Fig. 5.

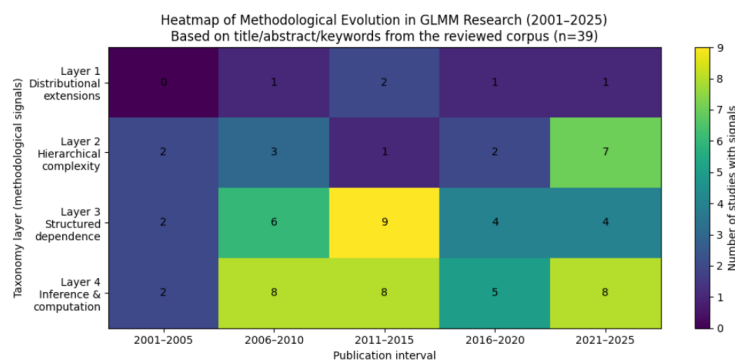


Fig. 5. Heatmap of methodological evolution in GLMM research (2001–2025)

The numbers displayed in each cell represent the number of studies in the reviewed corpus that contain methodological signals corresponding to the respective layer within the specified publication interval. These signals were identified through keywords and methodological descriptors reported in article titles, abstracts, and indexed keywords.

The heatmap reveals that methodological innovation in GLMM research has not occurred uniformly across layers. While the probabilistic and basic hierarchical components remain relatively stable over time, substantial growth appears in the modeling of structured dependence and computational inference strategies. This pattern reflects the increasing importance of spatial and spatio-temporal modeling as well as advances in computational statistics that enable more complex hierarchical models to be estimated. This pattern reflects the increasing importance of spatial and spatio-temporal modeling as well as advances in computational statistics that enable more complex hierarchical models to be estimated.

To provide a clearer illustration of how individual studies correspond to the structural components of the proposed taxonomy, selected representative studies were mapped to the four layers of the GLMM framework. This mapping highlights how specific methodological contributions relate to distributional assumptions, hierarchical structures, dependence modeling, and inferential strategies. The representative mapping of these studies is presented in Table A2 (Appendix A).

3.4 Cross-Layer Variance Decomposition and Identifiability

The four-layer taxonomy introduced in this study serves not only as a framework for organizing the structural components of generalized linear mixed models, but also as a lens for understanding how variability is generated and interpreted within these models. When GLMMs are viewed through this layered architecture, variance is no longer associated with a single parameter or model component. Instead, it emerges from a sequence of modeling decisions that operate at different stages of model specification. This perspective is consistent with the broader view of mixed models as hierarchical systems in which variability is distributed across multiple levels of the model structure (McCulloch et al., 2008; Jiang, 2007).

At the probabilistic layer (Layer 1), variability originates from the assumed conditional distribution of the response variable. The selected distribution determines the fundamental relationship between the mean and variance of the outcome. For instance, Poisson models impose equality between the mean and variance, whereas Negative Binomial formulations introduce an additional dispersion parameter that allows greater variability. In applied settings, this distributional choice establishes the baseline stochastic behavior of the data before hierarchical effects are introduced. Once dispersion parameters are specified at this stage, they inevitably interact with variance components defined in subsequent layers of the model (Lee et al., 2006).

The hierarchical layer (Layer 2) introduces additional variability through random effects that capture latent heterogeneity across clusters or observational units. From a conceptual standpoint, total variability can be interpreted as arising from two sources: variation inherent in the response distribution and variation attributable to differences between groups. Although this decomposition is theoretically straightforward, empirical estimation can become challenging. When cluster sizes are small, or when dispersion parameters and random-effect variances are estimated simultaneously, the model may struggle to uniquely attribute excess variability to a particular source. In such situations, variance components may approach boundary values or exhibit instability across estimation procedures, a phenomenon discussed in the literature on variance component estimation in mixed models (Sinharay & Stern, 2003).

Further complexity arises at the structured dependence layer (Layer 3). Spatial or temporal structures introduce correlation patterns that extend beyond simple clustering. Rather than representing independent group-level heterogeneity, variability becomes structured according to spatial proximity or temporal continuity. This structural constraint alters the interpretation of variance components. In particular, spatial random effects may partially overlap with fixed-effect covariates that exhibit similar geographic patterns, making it difficult to distinguish systematic associations from correlated residual variation. This issue, commonly referred to as spatial confounding, has been widely discussed in the context of spatial generalized linear mixed models (Hughes & Haran, 2013; Goicoa et al., 2018).

The inferential layer (Layer 4) ultimately determines how these sources of variability are quantified. Because the marginal likelihood of GLMMs typically requires integration over latent random effects, closed-form solutions are rarely available. As a result, estimation relies on approximation or simulation-based procedures such as Laplace approximation, penalized quasi-likelihood, or Bayesian computational approaches (Breslow & Clayton, 1993; Rue et al., 2009). The choice of estimation strategy affects how uncertainty is represented and how sensitive the results are to structural complexity. In models that incorporate high-dimensional spatial or spatio-temporal dependence, computational considerations often become closely intertwined with substantive interpretation (Lindgren et al., 2011).

Viewed through this layered perspective, variance in GLMMs should therefore be interpreted as a distributed property of the model rather than as a single estimable quantity. Identifiability issues frequently arise not from isolated modeling decisions, but from the interaction between distributional dispersion, hierarchical heterogeneity, structured dependence, and computational approximation. Recognizing these cross-layer interactions helps explain why variance components may behave differently across empirical applications and highlights the importance of considering model architecture as an integrated system when interpreting GLMM results.

While Fig. 5 illustrates how methodological developments in GLMM research have evolved across the four layers over time, it does not indicate how these components tend to appear together within individual studies. To address this structural aspect, Fig. 6 maps the reviewed studies according to the combinations of taxonomy layers they simultaneously incorporate.

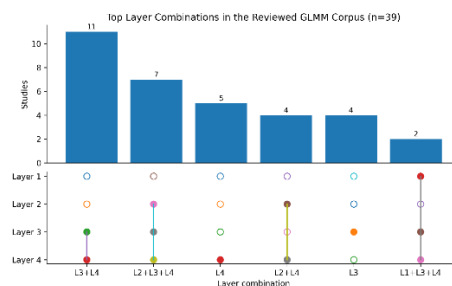


Fig. 6. Mapping of the reviewed studies onto the four-layer GLMM taxonomy (top six layer combinations, $n = 39$)

In Fig. 6, labels such as L2+L3 or L3+L4 denote combinations of methodological components that appear simultaneously within individual studies. The notation follows the four-layer taxonomy introduced earlier: L1 refers to the probabilistic likelihood specification, L2 represents hierarchical model structures with random effects, L3 captures structured dependence such as spatial or temporal correlation, and L4 denotes the inferential and computational strategies used for parameter estimation.

Under this notation, combinations such as L2+L3 indicate studies that integrate hierarchical random-effects structures with structured dependence components, whereas L3+L4 represents models that combine spatial or temporal dependence modeling with advanced computational inference.

The distribution of these combinations reveals several dominant methodological configurations within the reviewed corpus. The most frequent pattern corresponds to studies combining structured dependence modeling with advanced inferential strategies (L3+L4), suggesting that methodological developments in GLMM research are often driven by the need to represent spatial or spatio-temporal correlation while simultaneously adopting computationally intensive estimation techniques.

Another prominent configuration involves the joint presence of hierarchical random effects, structured dependence, and advanced computational inference (L2+L3+L4). This pattern reflects the increasing structural complexity of contemporary GLMM applications, particularly in epidemiological and spatial disease modeling contexts where multilevel structures and correlated observations frequently occur simultaneously.

Overall, the figure highlights how methodological innovation in the GLMM literature rarely occurs within a single modeling component. Instead, developments typically arise through the interaction of multiple layers of model construction, reinforcing the usefulness of the proposed taxonomy for interpreting the evolution of GLMM methodology. These patterns provide additional empirical support for the layered interpretation of GLMM methodology developed in this review and form the basis for the broader methodological discussion presented in the following section.

4. Discussion

This review examined generalized linear mixed models (GLMMs) not from the perspective of application domains, but from the standpoint of internal model construction. By shifting attention to structural components rather than subject matter, several regularities become apparent. Differences across studies are less about isolated modeling preferences and more about how layers of model structure are assembled and combined. The four-layer taxonomy proposed in this study provides a coherent framework for interpreting these patterns.

Beyond serving as a classificatory framework, the taxonomy can also be interpreted as a conceptual workflow describing how generalized linear mixed models are progressively constructed through successive modeling layers. Figure 7 illustrates this workflow perspective. The modeling process begins with observational data, followed by the specification of the probabilistic likelihood (Layer 1). Hierarchical random effects are then introduced to capture latent heterogeneity across clusters (Layer 2). Structured dependence components, including spatial or temporal correlation, may subsequently be incorporated (Layer 3). Finally, computational and inferential procedures are applied to estimate model parameters (Layer 4). Viewed in this way, methodological developments in the GLMM literature correspond to structural modifications introduced at different stages of model construction.

One notable observation concerns the relative stability of lower-layer modeling decisions. Most studies included in the review rely on standard exponential-family distributions together with relatively simple hierarchical formulations, typically random intercept structures. These elements appear to form the methodological backbone of applied GLMM research. In contrast, greater diversity emerges in the modeling of structured dependence and in computational strategies. Spatial, temporal, and spatio-temporal extensions frequently introduce substantial additional complexity, and such complexity often coincides with shifts in estimation paradigms.

The layered perspective also clarifies recurring challenges in variance interpretation. Variability in GLMMs is distributed across multiple components: distributional dispersion, cluster-level heterogeneity, and structured correlation. While these elements are conceptually distinct, empirical separation is not always straightforward. When several variance-generating mechanisms operate simultaneously, parameters may compete to explain similar patterns in the data. The resulting instability is therefore not necessarily a flaw of estimation, but rather a reflection of limited information available to disentangle overlapping sources of variability.

Structured dependence models further illustrate this point. Once correlation is encoded through spatial or temporal precision structures, heterogeneity is no longer exchangeable; instead, variability becomes partially constrained by structural assumptions. In such settings, the interpretation of regression coefficients and variance components depends critically on how the dependence structure interacts with fixed effects. These interactions highlight the importance of viewing identifiability not only as a numerical issue but also as a structural property of the model.

The inferential layer introduces an additional dimension to this interpretation. Because GLMM estimation typically requires integration over latent random effects, computational methods inevitably influence how uncertainty is represented. Approximation-based approaches and simulation-based procedures may display different practical behavior, particularly in models involving high-dimensional random effects or complex dependence structures. Computational considerations are therefore not external to model formulation; rather, they form an integral part of the modeling architecture.

Several limitations should be acknowledged. First, the review was restricted to studies retrieved from a single bibliographic database. Second, the analysis relies on reported model specifications rather than on re-estimation under a unified analytical framework. Consequently, observations regarding identifiability and variance behavior remain interpretative and are grounded in methodological descriptions rather than direct experimental comparison.

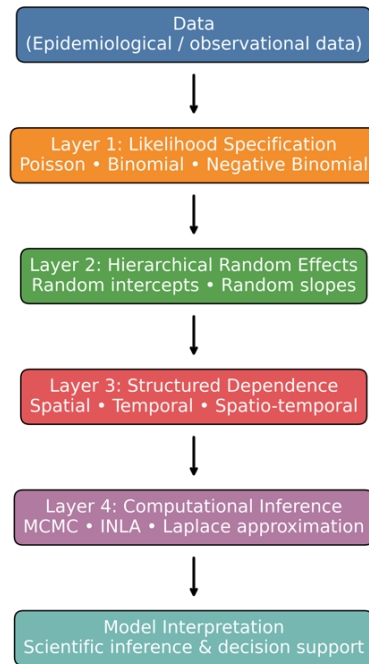


Fig. 7. Conceptual workflow of generalized linear mixed model construction across the four-layer taxonomy

Despite these limitations, organizing GLMM methodology into hierarchically connected layers provides a useful analytical lens. It highlights how model complexity accumulates through structural additions and how variance behavior emerges from interactions among multiple model components rather than from isolated parameters. This perspective may encourage clearer methodological reporting and support more explicit reflection on how modeling decisions at one structural level influence interpretation at another.

5. Conclusions

This review examined generalized linear mixed models (GLMMs) from the perspective of internal model construction rather than application domain. By shifting attention toward the structural components that define GLMM formulations, several regularities become visible across the literature. Differences between studies appear less as isolated methodological choices and more as combinations of structural elements introduced at different stages of model construction. The four-layer taxonomy proposed in this study provides a coherent framework for interpreting these patterns.

One of the most consistent observations concerns the relative stability of the lower layers of the model structure. Across the reviewed studies, most GLMM applications rely on standard exponential-family distributions and relatively simple hierarchical specifications, particularly random-intercept formulations. These components appear to function as the methodological backbone of applied GLMM research. In contrast, greater diversity emerges in the upper layers of the model structure. Extensions involving spatial, temporal, or spatio-temporal dependence frequently introduce substantial complexity, and these extensions are often accompanied by shifts in estimation strategy.

Viewing GLMM development through a layered perspective also clarifies how model complexity accumulates. Rather than emerging from entirely new modeling paradigms, many methodological innovations arise from structural modifications introduced within existing components of the model. For example, incorporating spatial dependence through conditional autoregressive structures, or introducing Bayesian estimation techniques such as MCMC or INLA, can substantially alter model behavior without changing the fundamental probabilistic formulation. This observation suggests that the evolution of

GLMM methodology has largely proceeded through incremental structural extensions rather than through radical conceptual reformulation.

The layered framework also helps explain recurring challenges in interpreting variance components. In GLMMs, variability is distributed across multiple elements of the model, including distributional dispersion, hierarchical heterogeneity, and structured dependence. While these components are conceptually distinct, their empirical separation may be difficult in practice. When several variance-generating mechanisms operate simultaneously, parameters may compete in explaining similar patterns in the data. The resulting instability should therefore not always be interpreted as a failure of estimation, but rather as a reflection of the limited information available to disentangle overlapping sources of variability.

Structured dependence models illustrate this issue particularly clearly. Once spatial or temporal correlation structures are introduced, heterogeneity is no longer exchangeable across observational units. Instead, variability becomes constrained by the assumed dependence structure. In such settings, regression coefficients and variance components may become sensitive to the interaction between fixed effects and correlated random effects. This highlights the importance of considering identifiability not only as a numerical property of estimation procedures but also as a structural characteristic of the model architecture itself.

Another important implication concerns the role of computation in modern GLMM methodology. Because estimation requires integration over latent random effects, computational methods inevitably shape how uncertainty is quantified and how models behave in practice. Approximation-based approaches and simulation-based procedures may lead to different inferential properties, particularly in models with high-dimensional random effects or complex dependence structures. Consequently, computational choices should be viewed not as purely technical decisions but as integral components of the modeling framework.

Several limitations should be acknowledged. First, the review was restricted to studies retrieved from a single bibliographic database, which may not capture the entire range of GLMM-related methodological developments. Second, the analysis relied on reported model specifications rather than on re-estimation of models under a unified analytical framework. As a result, observations regarding variance interpretation and identifiability are based on methodological descriptions rather than direct empirical comparison.

Despite these limitations, organizing GLMM methodology into hierarchically connected layers offers a useful analytical perspective. The taxonomy emphasizes that model complexity accumulates through structural additions and that variance behavior often reflects interactions between multiple components of the model. By making these structural relationships explicit, the proposed framework may support clearer methodological reporting and encourage more systematic reflection on how modeling decisions introduced at one layer influence interpretation at others.

This review explored the development of generalized linear mixed models (GLMMs) from the perspective of internal model construction rather than application domain. By synthesizing 39 studies published between 2001 and 2025, the analysis revealed a recurring structural pattern in how GLMM methodology has evolved across the literature. Differences between studies rarely reflect entirely new modeling paradigms. Instead, most developments arise from modifications introduced within specific components of an underlying model architecture.

To clarify these patterns, this study proposed a unified four-layer mathematical taxonomy that organizes GLMM methodology according to four structural components: probabilistic specification, hierarchical random effects, structured dependence, and inferential strategy. When the reviewed studies are interpreted through this framework, methodological variation becomes easier to understand. Lower layers of the model, particularly the choice of exponential-family distributions and basic hierarchical structures remain relatively stable across applications. Greater diversity appears in the upper layers, where spatial or temporal dependence structures and advanced computational estimation techniques are introduced.

Viewing GLMM development through a layered perspective also helps explain how model complexity accumulates. Many methodological innovations do not replace the underlying GLMM formulation but extend it by adding new structural elements. For example, spatial dependence models based on conditional autoregressive structures or spatio-temporal processes can substantially change model behavior while leaving the probabilistic foundation unchanged. Similarly, the adoption of Bayesian computational approaches such as MCMC or INLA alters the inferential layer without fundamentally modifying the hierarchical representation of the model.

The layered framework also provides insight into recurring challenges in variance interpretation. In GLMMs, variability may originate from multiple sources, including distributional dispersion, hierarchical heterogeneity, and structured correlation. Although these components are conceptually distinct, their empirical separation is often difficult when several variance-generating mechanisms operate simultaneously. Instability in variance components therefore reflects not only estimation difficulties but also structural interactions within the model itself.

Several limitations should be acknowledged. The review relied on studies retrieved from a single bibliographic database and was based on reported model specifications rather than re-estimation under a unified analytical framework. Consequently, the conclusions drawn here are interpretative and focus on methodological patterns rather than direct empirical comparison.

Despite these limitations, organizing GLMM methodology into hierarchically connected layers offers a useful analytical perspective. The proposed taxonomy highlights how methodological innovation frequently emerges from interactions between different structural components of the model. By making these relationships explicit, the framework may support clearer reporting of GLMM specifications, facilitate comparison between alternative modeling strategies, and provide a conceptual foundation for future methodological developments.

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Author contributions

RA and MAI conceived the study; RA and SW designed the study. RA collected and analyzed the data. RA, MAI and NC interpreted the data. RA and NC drafted the manuscript, and RA and MAI critically reviewed the manuscript. All authors read and approved the final manuscript.

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Data availability

All data included in this study are reported in the article.

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Appendix

Appendix A. Summary of Included Studies

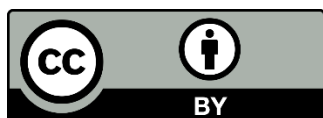
Table A1
Summary of Included Studies

No	Study	Year	Title (Shortened)	Journal
1	Yau	2001	Multilevel models for survival analysis with random effects	Biometrics
2	MacNab & Dean	2002	Spatio-temporal modelling of disease rates	Statistics in Medicine
3	Diggle et al.	2002	Model-based geostatistics for childhood malaria	JRSS C
4	Sinharay & Stern	2003	Variance component testing in GLMMs	ETS Research Report
5	Armero et al.	2008	Bayesian assessment of diagnosis times	Journal of Applied Statistics
6	Escaramis et al.	2008	Spatial conditional autoregressive disease risk detection	Biometrics
7	MacNab & Lin	2009	Empirical Bayes PQL inference in GLMMs	CSDA
8	Chu et al.	2009	Random effects models in diagnostic test meta-analysis	JASA
9	Hartman et al.	2009	Spatial smoothing for genetic mapping	CSDA
10	Lai & Yau	2009	Multilevel mixture cure models	Biometrical Journal
11	Torabi & Rosychuk	2010	Spatio-temporal disease mapping of rates	Canadian Journal of Statistics
12	Escaramis et al.	2011	Spatio-temporal child mortality analysis	Int. Journal of Health Geographics
13	Lowe et al.	2011	Climate-sensitive disease modeling for dengue	Computers & Geosciences
14	Torabi & Rosychuk	2011	B-spline disease mapping models	Journal of Applied Statistics
15	Hund et al.	2012	Geostatistical disease mapping	Biometrics
16	Hughes & Haran	2013	Spatial confounding in GLMMs	JRSS B
17	Lowe et al.	2013	Early warning system for dengue epidemics	Statistics in Medicine
18	Lekdee & Ingsrisawang	2013	Spatial random effects GLMM for dengue	Journal of Mathematics and Statistics
19	Graham et al.	2013	Longitudinal mixed models for injury risk	JRSS A
20	Yoo	2014	Prediction of mosquito abundance using GLMM	IJGIS
21	Hefley et al.	2017	Dynamic spatio-temporal spatial models	Spatial Statistics
22	Hefley et al.	2017	Bayesian group lasso for spatial data	Journal of Agricultural, Biological and Environmental Statistics
23	Lázaro et al.	2017	Bayesian correlated virus prevalence models	SORT
24	Goicoa et al.	2018	Identifiability constraints in disease mapping	SERER
25	Nikolouloupoulos	2019	D-vine copula mixed models	Statistical Methods in Medical Research
26	Tawiah et al.	2020	Zero-inflated hierarchical Poisson models	IJMMS
27	Abd Naeem et al.	2020	Spatio-temporal dengue modeling	Journal of Applied Statistics
28	Rashid et al.	2020	Modeling between-study heterogeneity	JASA
29	Lubinda et al.	2021	Spatio-temporal malaria risk modeling	PLoS Computational Biology
30	Mindra Jaya et al.	2021	Dengue transmission modeling	CMBN
31	Abd Naeem et al.	2022	Spatio-temporal dengue modeling	Bulletin MMS
32	Krause et al.	2022	Penalized likelihood inference in mixed models	Metrika
33	Jones et al.	2023	Chronic disease stepped wedge trial	Digital Health
34	Nikolouloupoulos	2024	Joint diagnostic test meta-analysis	Statistical Methods in Medical Research
35	MacDonald & Lee	2024	Basis representations for spatial data	Spatial Statistics
36	McMahan et al.	2025	Bayesian mixed-effects regression for group testing	Biometrics
37	Yilema	2025	Bayesian small-area anemia estimation	Statistical Journal of IAOS
38	Ocaña-Riola	2007	Misuse of aggregated count data	Statistics in Medicine
39	MacNab & Gustafson	2007	Bayesian disease mapping smoothing	Statistics in Medicine

Table A2

Representative mapping of selected studies

Study	Layer 1 Distribution	Layer 2 Hierarchy	Layer 3 Dependence	Layer 4 Inference
MacNab & Dean (2002)	Poisson	Random intercept	Spatio-temporal	Likelihood
Diggle et al. (2002)	Poisson	Random effects	Spatial	Likelihood
Hughes & Haran (2013)	Exponential family	Random effects	Spatial	Bayesian
Lowe et al. (2013)	Poisson	Random intercept	Spatio-temporal	Likelihood
Lubinda et al. (2021)	Poisson	Random intercept	Spatio-temporal	Bayesian
Abd Naeem et al. (2020)	Poisson	Random intercept	Spatio-temporal	Bayesian
Tawiah et al. (2020)	Zero-inflated Poisson	Random intercept	None	Bayesian
Krause et al. (2022)	Binomial	Random intercept	None	Penalized likelihood
Nikoloulopoulos (2019)	Binomial	Random effects	Copula Dependence	Bayesian
McMahan et al. (2025)	Binomial	Random intercept	None	Bayesian
Yilema (2025)	Binomial	Random intercept	Spatial	Bayesian
MacDonald & Lee (2024)	Exponential family	Random effects	Spatial	Bayesian
Hefley et al. (2017)	Exponential family	Random effects	Spatio-temporal	Bayesian
Goicoa et al. (2018)	Poisson	Random intercept	Spatial	INLA / PQL
Torabi & Rosychuk (2010)	Poisson	Random intercept	Spatio-temporal	Likelihood
Hund et al. (2012)	Poisson	Random effects	Spatial	Likelihood



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RESTU ARISANTI <r.arisanti@unpad.ac.id>

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Title: Generalized Linear Mixed Models in Epidemiological Data Analysis: A Systematic Review of Methodological Developments

Author 1:

First Name: Restu
Last Name: Arisanti
Organization: Universitas Padjadjaran
Country: Indonesia
Email: r.arisanti@unpad.ac.id

Author 2:

First Name: Maizatul Akmar
Last Name: Ismail
Organization: Universiti Malaya
Country: Malaysia
Email: maizatul@um.edu.my

Author 3:

First Name: Sri
Last Name: Winarni
Organization: Universitas Padjadjaran
Country: Indonesia
Email: sri.winarni@unpad.ac.id

Author 4:

First Name: Nita
Last Name: Cahyani
Organization: Universitas Padjadjaran
Country: Indonesia
Email: nita.cahyani@unpad.ac.id

Contact Author: Author 1

Alternate Contact:

Topic(s):

- Big data analysis
- Data processing
- Data visualization
- Data analysis and parameter tuning

Keywords: Computational inference; Generalized linear mixed models; Hierarchical modeling; Methodological taxonomy; Spatial dependence

Abstract: Generalized linear mixed models (GLMMs) have become a fundamental statistical framework for analyzing hierarchical and correlated data in epidemiology and related quantitative fields. By combining non-Gaussian response distributions with random effects and structured dependence, GLMMs allow researchers to represent complex data structures arising from spatial, temporal, and multilevel processes. Over the past two decades, numerous methodological extensions have been introduced, including developments in spatial modeling, hierarchical structures, and computational inference. Despite this growth, these contributions are often presented as separate methodological advances, making it difficult to understand how they relate within the broader architecture of the GLMM framework. This study addresses this issue through a systematic review of methodological developments in GLMM-based epidemiological research. Thirty-nine studies published between 2001 and 2025 were examined to identify recurring modeling patterns and structural extensions of the GLMM framework. Based on this synthesis, a four-layer mathematical taxonomy is proposed that organizes GLMM methodology according to probabilistic specification, hierarchical structure, structured dependence, and inferential strategy. The results indicate that most innovations arise from structural modifications within these layers rather than from entirely new modeling paradigms, providing a clearer perspective on methodological diversity in the GLMM literature.

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Dear *Restu Arisanti*

I would like to confirm that your paper entitled “**Generalized Linear Mixed Models in Epidemiological Data Analysis: A Systematic Review of Methodological Developments**” with *Maizatul Akmar Ismail, Sri Winarni and Nita Cahyani* has been accepted for publication on International Journal of Data and Network Science, An international journal.

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